

Short Paper: Non-Fundamental by Impossibility: A Public-Data Bound on Panic and Friction in the March 2023 USDC De-Peg

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Abstract. We give a reusable, public-data instrument for a disclosed-impairment de-peg. Under reserve-only fundamentals, par redemption when solvent, and recovery in $[0, 1]$ on the impaired fraction ϕ , fundamental value cannot fall below $1 - \phi$. In March 2023, SVB’s failure trapped about 8% of USDC’s reserves and the coin fell to \$0.877, below the \$0.92 floor. Under those three conditions, the trough requires unremediated-loss probability $q^* = 1.54$, exceeding one. At the disclosed ϕ , at least 35% of the de-peg is non-fundamental on the composite feed and at least 19% on Kraken’s close. An ex-ante fragility ordering by reserve exposure matches the realized troughs. A specificity panel gives one true positive (USDC, 2023), one regulator-quantified, issuer-contested episode on which the bound returns no verdict (Tether, 2019), and one degenerate control, not counted (USDT, Terra episode, May 2022, $\phi = 0$).

Keywords: Stablecoins · Bank runs · Global games · De-peg · Financial stability · Market microstructure

1 Introduction

Stablecoins are becoming payment infrastructure [32], and the GENIUS Act gives them their first comprehensive U.S. federal framework. A fiat-backed stablecoin that trades below par admits two readings. Under *rational repricing*, an impaired reserve is worth less. Under *non-fundamental* repricing, a self-fulfilling coordination run and microstructure friction push the price below the floor a reserve loss can justify. No measurement of March 2023 cited in Section 2 forms that floor. The policy response to a mostly non-fundamental de-peg is backstop and redemption design, since a reserve-quality rule addresses only the fundamental share.

We offer a reusable *estimator* for any disclosed-impairment de-peg. From an issuer’s *disclosed* reserve impairment, derive a worst-case floor. Test whether the observed trough breaches that floor. Read the breach, if any, as a bound on the non-fundamental share. A supervisor can run it on public data within hours of a break. It returns no verdict when the trough does not breach the floor.

We apply it to the March 2023 de-peg of USD Coin (USDC). The failure of Silicon Valley Bank (SVB) trapped about 8% of USDC’s reserves. A joint federal backstop [5] restored the peg within days. The SVB failure was an idiosyncratic shock with a dated disclosure, and its reversal was complete. The counterfactual

“no panic” price is therefore pinned twice, as a worst-case floor and, ex post, as the restored peg.

Under expected-loss pricing, fundamental value cannot fall below $1 - \phi$, where ϕ is the disclosed impairment fraction. A preference-free corollary extends that floor to any monotone-utility valuation. The observed trough breaches the floor at every admissible unremediated-loss probability and recovery rate, and requires $q^* = 1.54$, which exceeds one. No rational-pricing account is consistent with the trough. The resulting non-fundamental share stays positive across impairment readings, and at the full re-peg it is 100% ex post, an identity (Section 4).

The paper makes two contributions:

- **A model-free impossibility bound** (Proposition 1) on what a disclosed reserve impairment can justify.
- **An offline, keyless reproduction path**: every computed number regenerates from a frozen snapshot with no network call and no API key.

Two analyses stay outside the estimator: the ex-ante fragility ordering by reserve exposure that the realized troughs match, with null probability 1/2 (Section 5), and a three-episode specificity panel (Section 6). We screen seven candidate de-pegs. In none did an issuer with a material impairment disclose its size while the exposure was unresolved, which limits qualifying episodes to what issuers choose to disclose.

2 Related Work

Diamond and Dybvig’s [13] demand-deposit model leaves the run *probability* undetermined, and global games select a unique equilibrium [7,28,18]. We take only the selection result, and our bound does not depend on uniqueness.

Since 2020 a literature applies run theory to stablecoin design and peg arbitrage [26,21,23,2]. Anadu et al. [3] analyze the March 2023 de-peg through a discrete “break-the-buck” threshold. Watsky et al. [34], and a Federal Reserve post-mortem they co-authored [15], attribute that threshold to a redemption window restricted to Circle’s approved customers over the March 10–13 weekend. Secondary prices de-pegged while the arbitrage loop was closed. Uhlig [31] finds many TerraUSD holders behaved as rational Bayesian updaters in May 2022, evidence that rational and panic behavior are hard to separate without a structural restriction.

Other work measures the March 2023 de-peg without an expected-loss floor: the SVB link [17], a daily cross-coin exposure ranking [14], a detection lead of about five hours before USDC crosses \$0.99 [8], a settlement-friction framing [22], and the on-chain contagion pathway [29]. Our bound derives an expected loss from disclosed exposure. None of the papers just cited does.

Contribution boundary. Structural calibrations [24] estimate a run *probability*, and event studies [3] document *flows*. We bound one episode’s realized non-fundamental share from an accounting identity. Anadu et al.’s [3] *rational* run

concerns the redemption decision and makes no claim about whether the price was justified. Watsky et al.’s [34] weekend discount is the microstructure friction the non-fundamental share includes. Ahmed, Aldasoro, and Duley [1] run synthetic-control tests on four *other* episodes and treat March 2023 only as a price path. Zhu [35] classifies *which* channel moves price and leaves *how far* unmeasured. Molinet and Filos-Ratsikas [27] define fiat-backed coins as the always-stable case ($1/p_t = 1$) and never analyze March 2023.

3 Setting and Data

USDC is a fiat-backed stablecoin issued by Circle, redeemable one-for-one against reserves held in short-dated U.S. Treasuries and bank cash, so a bank failure impairs backing through the cash leg. The peg depends on full reserve backing and on an arbitrage loop, in which an authorized participant can buy tokens below par and redeem them at \$1.

Silicon Valley Bank failed on March 10, 2023. Late that evening, Circle disclosed on its official X account that “\$3.3 billion of the ~\$40 billion of USDC reserves remain at SVB” [9]. Circle restated that figure as “about 8%” of total reserves in its March 12 press release [10], and contemporaneous reporting corroborates it [6]. We take $\phi \approx 0.08$, Circle’s own rounded disclosure. The secondary-market price fell below par over March 10–11, reaching a trough of \$0.877. On March 12 at 22:15 UTC, the FDIC, the Federal Reserve, and the Treasury issued a joint statement guaranteeing SVB depositors in full [5], and USDC restored its peg over the following days.

DAI, the *contagion* coin, is linked to USDC through MakerDAO’s peg-stability module, which swaps the two one-for-one up to a debt ceiling that press reporting says was reached on 11 March 2023 [12]. Tether (USDT), the largest fiat-backed stablecoin, had no disclosed exposure to SVB and serves as the *unexposed* coin.

All analysis runs offline on a frozen snapshot over the window 2023-03-08 to 2023-03-17 (UTC), and no proprietary data enters it. The headline USD price series is an hourly composite from the public aggregator DeFiLlama. The frozen record does not enumerate its constituent venues, so we do not treat it as an independent venue. The hourly low of Kraken’s fiat-settled USDC/USD pair, with a minimum print of \$0.874, cross-checks the \$0.877 trough.

4 The Bound and Its Application

The estimator is pure accounting, with no run threshold and no fitted parameter. Let ϕ be the fraction of reserves impaired and $\rho \in [0, 1]$ the expected recovery rate on the impaired assets. The fraction permanently lost is $\phi(1 - \rho)$, so expected backing per coin is

$$P_{\text{fund}} = 1 - \phi(1 - \rho), \quad (1)$$

and its minimum over ρ , the worst-case floor, is $1 - \phi$. The estimator rests on three assumptions, and a fourth serves the corollary below.

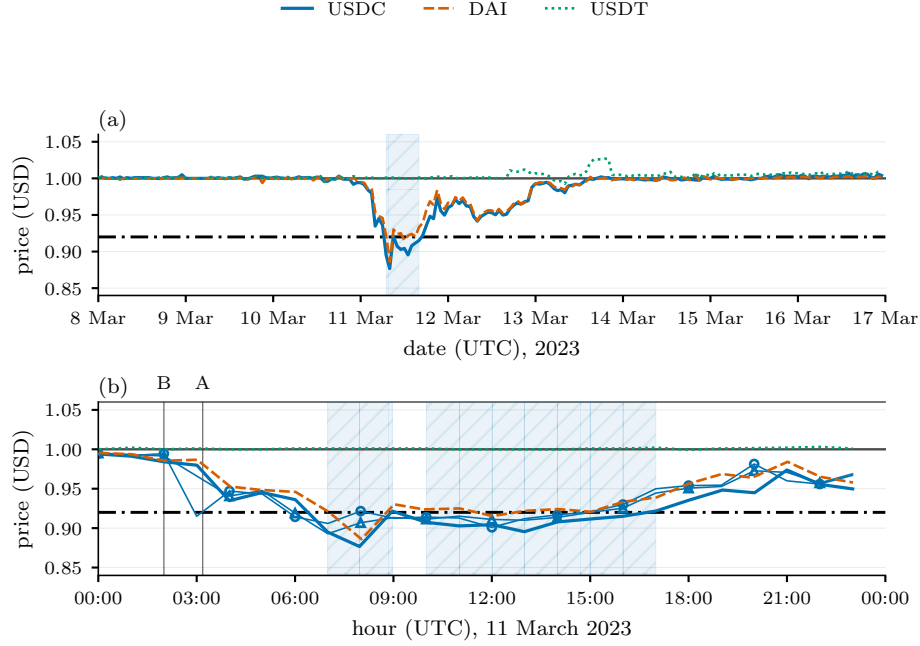


Fig. 1. Hourly USD prices of USDC (solid), DAI (dashed), and USDT (dotted) over the analysis window (a) and over 11 March 2023 (b), against par and the \$0.92 floor. Shading in (a) and hatching in (b) mark the impossibility window, 06:59–16:00 UTC on 11 March; nine of its ten hours breach the floor, and the tenth, at 08:57, recovers to \$0.9216 and is drawn unhatched. Circles in (b) are Kraken’s close and VWAP readings. Lines in (b) mark A, Circle’s disclosure (03:11 UTC), and B, the Coinbase USDT-USD premium peak (02:00 UTC), which fall 71 minutes apart and are not separable in (a).

- (A1) **Reserve-only fundamental.** A coin’s fundamental value is its expected reserve backing per unit given *disclosed* reserves, and the only impairment is the exposure ϕ .
- (A2) **Par redemption when solvent.** Conditional on rescue, redemption is at \$1 within the analysis horizon.
- (A3) **Bounded recovery.** Recovery on impaired reserves satisfies $\rho \in [0, 1]$, because the coin is fully reserved and the reserves carry no leverage.
- (A4) **Monotone-utility valuation.** The unconstrained marginal holder values the claim as an expected utility of its payoff, for some monotone increasing utility and some probability measure over outcomes, with no ambiguity about the disclosed ϕ . This nests risk neutrality, any degree of risk aversion, and rank-dependent weighting.

Proposition 1 (Rational-pricing floor and impossibility). *Assume (A1)–(A3), and let the impaired fraction’s loss go unremediated with probability $q \in [0, 1]$ under the pricing measure, so that $P_{\text{fund}}(q, \rho) = 1 - q\phi(1 - \rho)$. Then*

$P_{\text{fund}}(q, \rho) \in [1 - \phi, 1]$ for all $q, \rho \in [0, 1]$, so any observed price $P_{\text{obs}} < 1 - \phi$ cannot be rationalized as expected loss. The unremediated-loss probability it would require, at the worst-case recovery $\rho = 0$, is $q^* = (1 - P_{\text{obs}})/\phi > 1$.

Proof. P_{fund} is decreasing in q and increasing in ρ , so it is minimized at $(q, \rho) = (1, 0)$ at value $1 - \phi$ and maximized at $q = 0$ at value 1, and q^* solves $P_{\text{obs}} = 1 - q\phi$ at $\rho = 0$. $\square$

This is the floor-at-recovery-value bound of credit pricing, under which risky debt is bounded below by its recovery value for any pricing measure [25], with the unremediated-loss probability a reduced-form failure intensity [16].

Under (A1)–(A4) the floor is also preference-free. By (A2) and (A3) the payoff X on a USDC claim has support in $[1 - \phi, 1]$ in every state. For any monotone increasing utility u , the certainty equivalent $u^{-1}(\mathbb{E}[u(X)])$ of a lottery with that support lies in $[1 - \phi, 1]$ as well. That floor bounds rational *valuations*, meaning willingness to pay. It does not bound market-clearing *prices*, because a seller under binding margin or capital constraints can rationally transact below every unconstrained valuation [30,20]. That channel is a friction, and the non-fundamental share defined next includes it. The floor is as strong as the support condition $X \in [1 - \phi, 1]$ and no stronger. That condition does not cover issuer-wrapper risk, an insolvency that delivers less than the floor or delivers it late through claim-priority uncertainty, administrative cost, or a gated distribution [19].

Write the observed deviation $1 - P_{\text{obs}}$ and the fundamental deviation $\phi(1 - \rho)$. The non-fundamental share of the de-peg is

$$s_{\text{nf}}(\rho) = \frac{(1 - P_{\text{obs}}) - \phi(1 - \rho)}{1 - P_{\text{obs}}}. \quad (2)$$

Its lower edge takes $\rho = 0$, treating total permanent loss of the impaired reserves as the fundamental cap. Its upper edge takes $\rho = 1$, the full re-peg, which implies zero permanent loss and gives $s_{\text{nf}}(1) = 1$, exactly 100% ex post, an identity. The residual covers coordination panic and microstructure friction together, so the share is an upper bound on pure coordination panic.

With the disclosed impairment $\phi = 0.08$, the worst-case floor is $1 - \phi = \$0.92$ and the observed trough is $P_{\text{obs}} = \$0.877$. The trough lies below the floor, so under (A1)–(A3) no expected loss rationalizes it, and the implied unremediated-loss probability is $q^* = (1 - 0.877)/0.08 = 1.54$. The lower edge of (2) is $(0.1233 - 0.08)/0.1233 = 0.351$, so at least 35% of the de-peg is non-fundamental, and we report the band [35%, 100%], whose upper end the full re-peg supports. Across three readings of Circle’s disclosure and five price-feed troughs the lower edge runs from 16.7% to 36.4%, and every one of those fifteen combinations breaches the floor. At the disclosed ϕ , the lowest is 19.2%, from Kraken’s close, whose minimum print is \$0.901. By Proposition 1 the trough is unrationalizable for any impairment below 12.3%, a 4.3-point margin over the disclosed 8%, and the lower edge stays strictly positive across that margin.

(A1) binds, so ϕ is Circle’s disclosed figure and we report a band around it. Evaluating at $\rho = 0$ is the conservative choice, because it uses the lowest floor

that (A3) allows. Lowering the floor further would require relaxing (A3) below zero. Supply at the trough exceeded the \$40B denominator Circle used, so ϕ on the actual reserve base is 0.079, below the headline 0.08. The count of breaching hours over the daily-supply interpolation band $\phi \in [0.078, 0.081]$ ranges from 9 to 11.

Circle’s disclosure quantifies only the SVB deposit. The same update describes the remaining cash reserve as held at a variety of institutions, of which SVB was one, without stating how that remainder splits across banks [11]. An undisclosed exposure at another bank is therefore not ruled out by the disclosure itself. That undisclosed remainder falls inside the 12.3% margin established above, so the argument survives unless it pushed the true impairment past that figure.

Kraken’s USDC/USD close breaches the \$0.92 floor in 9 of 240 hours and its VWAP in 10. The hourly low breaches it in 12, but the three extra hours rest on one print worth \$2,948.

Because the estimator needs only a price and a publicly quantified ϕ , it can be evaluated at every hour of the frozen window (Figure 1). Of USDC’s 212 in-window hourly readings, 9 lie below the \$0.92 floor, and all nine fall on 11 March 2023 between 06:59 and 16:00 UTC. Kraken’s 240 rows cover a wider 8 to 17 March span than the composite’s 212 in-window hours, which is why its count is read against 240 and USDC’s against 212. That impossibility window measures duration and specificity. It adds no evidence volume, because hourly prices within one episode are near-unit-root, with lag-1 autocorrelation $r = 0.963$, so the 212 readings carry an AR(1)-adjusted effective sample size $n(1 - r)/(1 + r) = 4.0$. The lower edge, q^* , and the floor-breach indicator are three readings of $1 - P_{\text{obs}} - \phi$, so their agreement corroborates nothing.

5 Cross-Coin Evidence

Reserve exposure orders the three coins ex ante: USDC held direct SVB exposure, DAI was pinned to USDC through the peg-stability module, and USDT held none. Global-games run models make that exposure the state that sets a coin’s run threshold [7,28,18,24], giving the ex-ante fragility ordering USDC, DAI, USDT. Troughs are \$0.877, \$0.886, and \$0.992 in that order. The deviations from par are 1233, 1141, and 76 bps. The prices are shown to three figures and the deviations are computed on the unrounded four-figure prints, so the two reconcile at full precision rather than at the precision shown. DAI’s mechanical link to USDC leaves one free comparison, so the realized ordering has probability 1/2 and illustrates the ex-ante ordering without testing it. USDC and DAI clear the 1% break threshold, a deviation of at least one cent from par, by factors of 12 and 11. USDT does not clear it, so the placebo criterion is satisfied: the exposed coins break and the unexposed coin does not.

USDT as the identifying placebo. USDT was bid to a premium while USDC and DAI fell. That pattern is consistent with the crypto safe-haven behavior documented for USDT [4]. On Coinbase, \$1.24B of \$1.52B in window volume traded

Table 1. Specificity panel. The USDC row reads a composite headline print and the Tether row a Kraken hourly low, a mismatch that runs against the panel here, because Tether fails to fire even on its most favorable reading. q^* is undefined where $\phi = 0$.

Episode	Disclosure	ϕ	Floor	Trough (feed)	q^*	Fires	Margin
USDC 2023	Mar [9], 11 Mar 03:11 UTC	0.08	0.92	composite, headline: \$0.877 (11 Mar 07:57 UTC)	1.54	Yes	-0.043
Tether [a]	2019 NYAG press release, 25 Apr 2019	[0.259, 0.332]	[0.668, 0.742]	Kraken, low: \$0.955 (26 Apr 2019 02:00 UTC)	[0.135, 0.174]	No verdict	[-0.214, 0.288]
USDT, Terra episode (May 2022) [b]	no disclosure	0.00	1.00	composite: \$0.993	-	Yes (not counted)	-0.007

[a] regulator-quantified, issuer-contested. [b] degenerate by construction, with $\phi = 0$, and not counted as a panel member.

above par, so the premium held on material volume. The premium is venue- and time-specific. On Coinbase USDT-USD it peaked at +266 bps at 02:00 UTC on 11 March, 71 minutes *before* Circle’s 03:11 UTC disclosure. The composite feed peaks later and higher, at +270 bps at 19:00 UTC on 13 March. That peak falls after the joint federal backstop of the previous evening, so it is post-backstop rather than intra-run. The Coinbase premium responds to the SVB failure itself, because Circle’s USDC-specific disclosure had not yet been made. In the 02:00 UTC hour, Kraken USDT/USD printed a +600-bps high on two trades totaling 225 USDT, against a volume-weighted average premium of +95 bps for that hour. A common-factor repricing of stablecoin risk would move all three coins down together, so no such repricing can produce the 02:00 UTC premium, which is a flight-to-safety signature.

6 Specificity

The estimator requires three preconditions. First (i), ϕ must be publicly quantified while the exposure is unresolved. Second (ii), the price must fall more than 1% below par while that disclosure is live. Third (iii), the fundamental channel must be reserve-only, so that $1 - \phi$ is the floor. We screened seven candidates: BUSD, Tether (2022), DAI (2020), USDR, USDP, GUSD, and TUSD. None satisfies all three. Each fails in one of two ways: no disclosed impairment, or a loss measured under a different valuation model. Issuers with a material impairment did not disclose its size in any of the seven screened cases, so one qualifying episode is a property of the disclosure record.

The panel gives one true positive in USDC (2023), one regulator-quantified, issuer-contested episode in Tether (2019) on which the bound returns no verdict, and one degenerate control in USDT, Terra episode (May 2022). Tether’s episode is assessed under precondition (i). The New York Attorney General’s 25 April 2019 statement quantified Tether’s exposure and Tether contested the characterization. The statement implies $\phi \in [0.259, 0.332]$. The deepest post-disclosure Kraken USDT/USD print, \$0.9551 on 26 April 2019, is 21.4 to 28.8 cents above every worst-case floor, and $q^* \in [0.135, 0.174]$ (Table 1). We read both the USDC row and the Tether row at zero recovery on the impaired reserves, so the same one-sided bound governs both. The Terra row carries $\phi = 0$, so its floor is par, q^* is undefined, and the row is not counted.

Read strictly, precondition (i) demands the issuer’s own public quantification while the exposure is unresolved, and under that reading the panel contains no validated true negative. One episode on which the bound returns no verdict does not measure a false-positive rate. The specificity of the bound is unmeasured.

TrueUSD carried an unremediated reserve impairment during our window that was not publicly quantified until a September 2024 SEC complaint [33], so the estimator would draw $\phi = 0$ from its contemporaneous attestations. That is the Terra row’s degenerate case, so the bound returns no verdict. We define the residual relative to what the market could know, because a concealed impairment produces no detectable signal by construction.

7 Limitations and Conclusion

The panic-versus-fundamentals split is contested [3,34]. The composite feed could mismeasure the trough. The disclosed ϕ could understate exposure, and the 12.3% margin answers that. A time-value discount prices a delayed redemption for a holder who has an approved claim. Convertibility risk is the cost to a holder without an approved claim during the restricted window. That cost is not bounded by the discount and belongs to the residual. No public data source can bound it further. We identify the mechanism from a single episode. The ex-post 100% is an aggregate statement, and holders who sold at the trough still realized losses.

An anonymized artifact at <https://anonymous.4open.science/r/usdc-depeg-artifact-F62D/> carries the frozen snapshot, the seeded analysis code, and a manifest with a SHA-256 digest for every file it lists. We audited its commit metadata, README and license lines, and manifest paths for identifying information. Every computed number and figure regenerates offline from the artifact without an API key.

A disclosed impairment implies a worst-case floor, and the USDC trough lies below it. Under (A1)–(A3) and at the disclosed ϕ , at least 35% of the de-peg was non-fundamental by impossibility on the composite feed, and at least 19% on Kraken’s close. A mostly non-fundamental de-peg argues for backstop and redemption design. The estimator needs only a price and a publicly quantified ϕ .

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